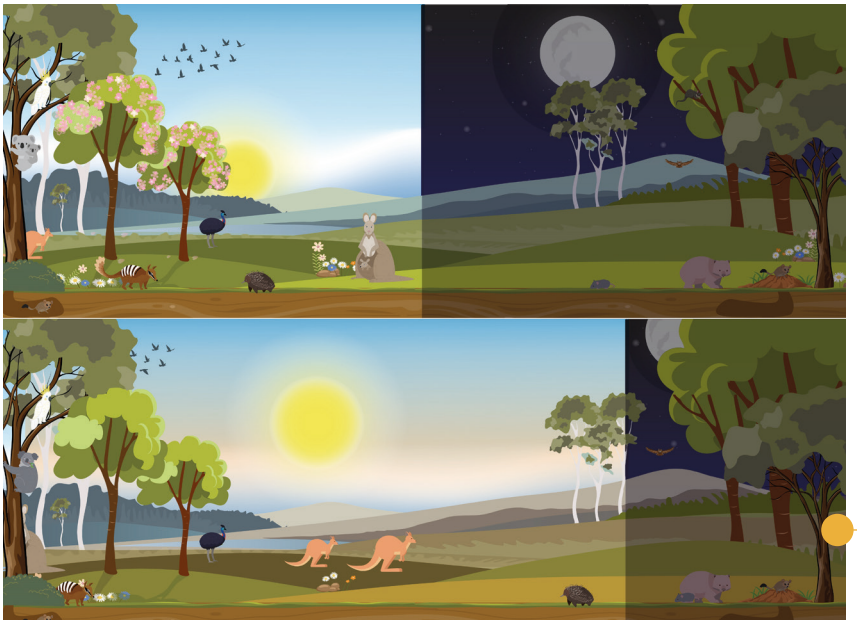
National Light Pollution Guidelines for Wildlife

What is light pollution?

Artificial light helps humans work, play and travel safely at night. However, artificial lighting that is inappropriate, excessive or poorly designed spreads into unwanted places. This is known as light pollution, and it disrupts the health and wellbeing of humans and wildlife alike.



How does wildlife use natural light?

Animals and plants use natural light signals from the sun, moon and stars to time their behaviour and life processes such as:

- patterns of rest and activity
- growth, reproduction and migration
- navigation over short and long distances.

Light pollution harms wildlife and ecosystems

Light pollution can mimic, mask or confuse natural light signals, causing:

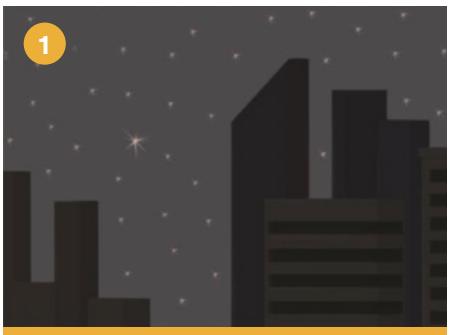
- mistimed activity, growth or breeding
- disturbed sleep and circadian rhythms
- disorientation and poor navigation
- attraction to artificial lights
- encounters with new predators
- reduced survival and reproduction.

Artificial lighting affects whole ecosystems by:

- dividing and disconnecting suitable habitat
- reducing pollination by nocturnal animals
- disrupting food webs and nutrient webs
- benefiting invasive species (cats, foxes and cane toads take advantage of artificial lights to feed).

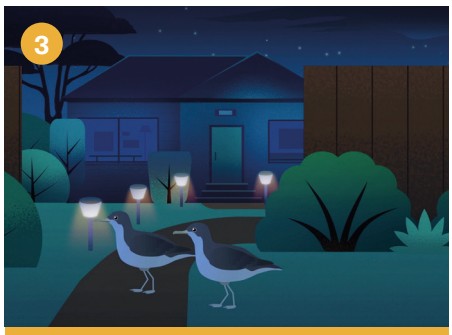
Reducing the Effects of Light Pollution on Wildlife

Use the **6 Best Practice Lighting Design Principles** to protect and restore natural darkness with good lighting design.



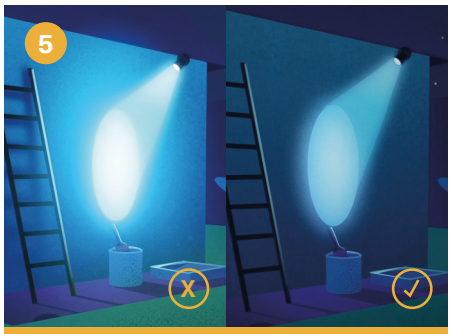
Start with natural darkness and only add light for a specific purpose.

Use adaptive light controls to manage the timing, intensity and colour of light.

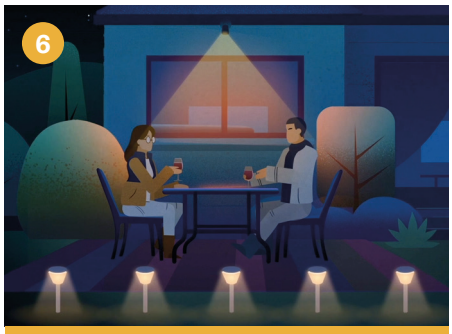


Light only the area needed.

Use low-intensity lighting and keep it close to the ground.



Use non-reflective, dark-coloured surfaces near lighting fixtures.



Avoid white lights - use amber (low-CCT) lighting with little or no blue wavelength.

Wildlife are highly sensitive to light

- Many animals are instinctively drawn towards light
- Animals respond to extremely faint light signals (including moonlight and even starlight)
- Light sources can affect animals many kilometres away, or deep under water.

We don't all experience light the same way

- Light that appears dim to humans may be bright for some wildlife
- Plants are also sensitive to light pollution
- Birds and bats view lights from high above; fish, turtles and frogs from below.

Monitor the effects of lighting

- Where a project is within 20km of habitat or migration routes of threatened species, migratory species or threatened ecological communities, impact of light on wildlife should be considered in more depth.
- An environmental impact assessment of artificial lighting (including mobile light sources such as vehicles and vessels) can identify potential light pollution and solutions.
- Once installed, artificial lighting and animal behaviour should be monitored to ensure light pollution has been sufficiently mitigated for wildlife.

For further information please scan the QR code or refer to the National Light Pollution Guidelines for Wildlife which can be downloaded at:
dcceew.gov.au/environment/biodiversity/publications/national-light-pollution-guidelines-wildlife

Australian Government
Department of Climate Change, Energy, the Environment and Water